Genetic Algorithm for Optimization: 09/10/2026

Algorithm Genetic Optimization

Input: Pop-size, cross_over_Rate, Mutation_Rate, Max_Gen

Output: best_solution

Define fitness (individual)

population ← random individuals (Pop_size)

for generation = 1 to MAX_GEN do

Evaluate fitness of every individual

Update best_solution if better one is found

new_population ← empty

while size(new_population) < Pop_size do

parent1, parent2 ← select (population)

~~if random() < crossover(parent1, parent2~~

if random() < crossover_rate then

child1, child2 ← ~~parent1, parent2~~

crossover (parent1, parent2

else

child1, child2 ← parent1, parent2

Mutate(child1), Mutate(child2) with

Mutation_rate

Add child1, child2 to new_population

END While

population ← new_population

END for

RETURN best_solution

END

Wireless Sensor Network Optimization

Pr: select cluster head & communication Routs for Sensor Nodes.

Obj: Minimize Energy Consumption & Extend Net work lifetime

Algorithm Genetic_Optimization

Input:
N = number of Nodes
E[i] = initial energy of node i
D[i][j] = distance B/w nodes i and j
P = Population size
G = number of Generations

Output: Best Cluster Head And Communication Routes

Initialize population with P random solutions

For each solution:
Select Cluster heads
Assign wireless Nodes to cluster heads
Select Communication Routes

calculate the total energy consumption
calculate network lifetime
calculate fitness of each solution
Update best_solution if better one is found
while size(new_population) < G do
network1, network2 ← ~~CROSSOVER~~ Select (population)
if random() < crossover_Rate then
child1, child2 ← crossover (Network1, Network2)
ELSE
child1, child2 ← Network1, Network2
END IF

MUTATE (child1), MUTATE (child2)

~~Add child1, child2~~

new_population ← child1, child2

END WHILE

population ← new_population

END FOR

RETURN BEST_NETWORK

END

Objective: Minimize Energy Consumption And Extend Network Lifetime By Optimally selecting Cluster Heads And Communication Routes In A Wireless Network Using A Genetic Algorithm.

Output:

=== Optimization Summary ===

Optimal cluster Heads: [5, 8]

Best Fitness Score : 178.5429

=== Optimal Routing Map ===

Node 0 connects to → cluster Head 5 | Distance : 27.25m

Node 1 connects to → cluster Head 8 | Distance : 64.16 m

Node 2 connects to → cluster Head 8 | Distance : 42.13 m

Node 3 connects to → cluster Head 8 | Distance : 21.96 m

Node 4 connects to → cluster Head 5 | Distance : 0.16 m

Node 5 is designated as [CLUSTER HEAD]

Node 6 connects to → cluster Head 5 | Distance : 15.55 m

Node 7 connects to → cluster Head 8 | Distance : 28.08 m

Node 8 is designated as [CLUSTER HEAD]

Node 9 connects to → cluster Head 8 | Distance : 17.68 m

===ROUTE VALIDATION ===

All nodes are connected to valid cluster Heads.